🔑 **Answer Key — Teacher Copy · Lesson 3-10**

MSTAR PRACTICE WORKSHEET

Lesson 3-10: Convert Measurement Units

Grade 6 Mathematics · Standard 6.AT.3c

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: C

How many inches are in 7 feet?

A. 70 inches

B. 72 inches

C. 84 inches

D. 96 inches

7 × 12 = 84 inches.

Question 2 SELECTED RESPONSE — CORRECT: B

How many feet are in 3 yards?

A. 6 feet

B. 9 feet

C. 12 feet

D. 36 feet

1 yard = 3 feet. 3 × 3 = 9 feet.

Question 3 SELECTED RESPONSE — CORRECT: D

How many ounces are in 4 pounds?

A 16 ounces

B 32 ounces

C 48 ounces

D 64 ounces

1 pound = 16 ounces. 4 × 16 = 64 ounces.

Question 4 SELECTED RESPONSE — CORRECT: A

Convert 24 inches to feet.

A 2 feet

B 3 feet

C 12 feet

D 24 feet

24 ÷ 12 = 2 feet.

Question 5 SELECTED RESPONSE — CORRECT: B

A shelf is 30 inches long. How many feet and inches is that?

A 1 foot 18 inches

B 2 feet 6 inches

C 2 feet 8 inches

D 3 feet 0 inches

30 ÷ 12 = 2 remainder 6. So 30 inches = 2 feet 6 inches.

Question 6 SELECTED RESPONSE — CORRECT: D

Leg 1: The joystick rod is 3 feet long, but the parts catalog lists it in inches. How many inches is that? (1 ft = 12 in)

A 15 inches

B 30 inches

C 33 inches

D 36 inches

Inches are smaller than feet, so multiply: 3 × 12 = 36 inches.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7 TWO-PART QUESTION

Part A — correct answer: D

Convert 3 hours to seconds using 1 hour = 60 minutes and 1 minute = 60 seconds.

A 120 seconds

B 180 seconds

C 3,600 seconds

D 10,800 seconds

3 hours × 60 minutes per hour = 180 minutes. Then 180 × 60 seconds per minute = 10,800 seconds.

- **A:** That multiplies only two of the three numbers. Go hours to minutes to seconds: 3 x 60 x 60.
- **B:** That is 3 hours converted to MINUTES. Convert once more: 180 minutes x 60 seconds per minute.
- **C:** That is the number of seconds in ONE hour. Multiply by 3 for three hours.

Part B — correct answer: D

In dimensional analysis, how are the conversion ratios arranged?

A So all units are multiplied together and kept.

B So the largest unit is always on top.

C So the numbers are as small as possible.

D So the unwanted units appear in both a numerator and a denominator and cancel.

Writing 3 hr × (60 min / 1 hr) × (60 s / 1 min) puts 'hr' and 'min' in both a numerator and a denominator, so they cancel and only seconds remain.

Question 8 SELECT ALL THAT APPLY — CORRECT: A, B, C, E

Select ALL statements that are true about dimensional analysis:

- **A** The given measurement is written as a fraction over 1.
- **B** Conversion ratios are arranged so unwanted units cancel.
- **C** Only the unit that remains after cancelling appears in the answer.
- **D** Units may be ignored as long as the numbers are multiplied correctly.
- **E** More than one conversion ratio can be used in a single chain.

A, B, C, and E are true. D is false — tracking the units is exactly what makes the method work and reveals a ratio written upside down.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9 WRITTEN RESPONSE · 2 POINTS

A recipe says you need 5 feet of ribbon, but the store only sells ribbon in inches. Another recipe says you need 2 yards of fabric, but your ruler measures in feet. Convert both measurements and explain the steps you used. Which conversion factor did you use each time?

Model answer: Both conversions go from a bigger unit to a smaller one, so both multiply. Using 12 inches per foot, 5 × 12 = 60 inches of ribbon; using 3 feet per yard, 2 × 3 = 6 feet of fabric. Naming the conversion factor each time is what explains why both numbers grew.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.